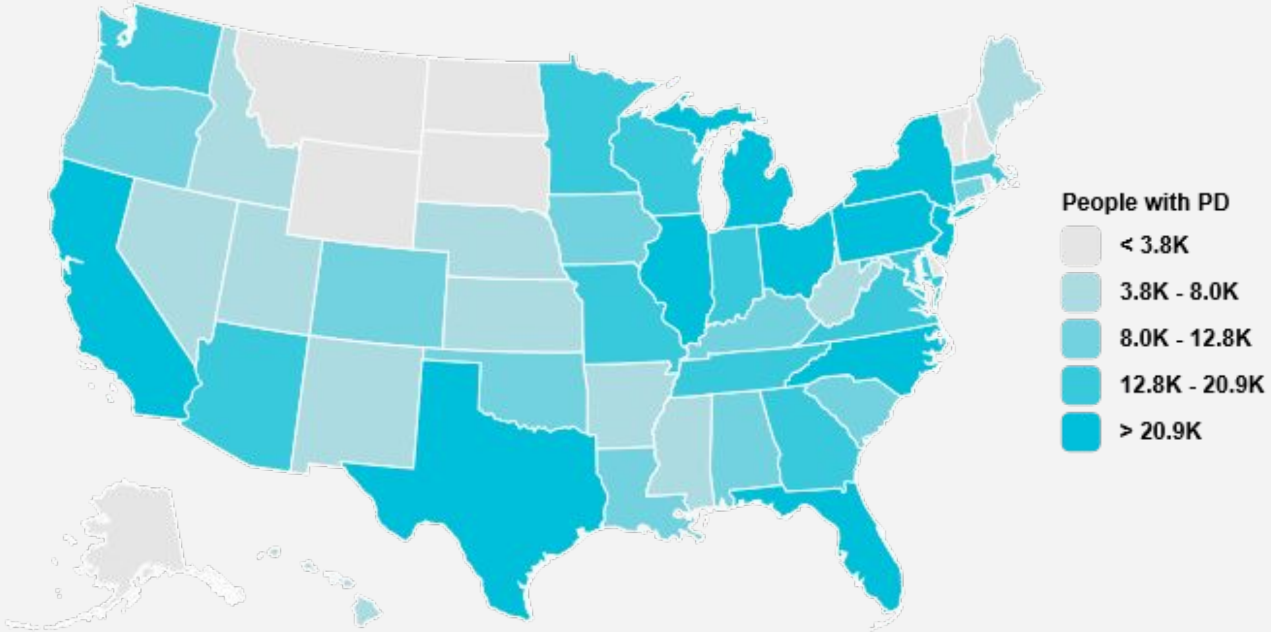
Do you know how many people are diagnosed with Parkinson's every year in the US?



Man-made measures:

Using AI to Augment Gene And Stem Cell Therapies to treat Parkinson’s Disease

Haashir Mohammed
Grade 10

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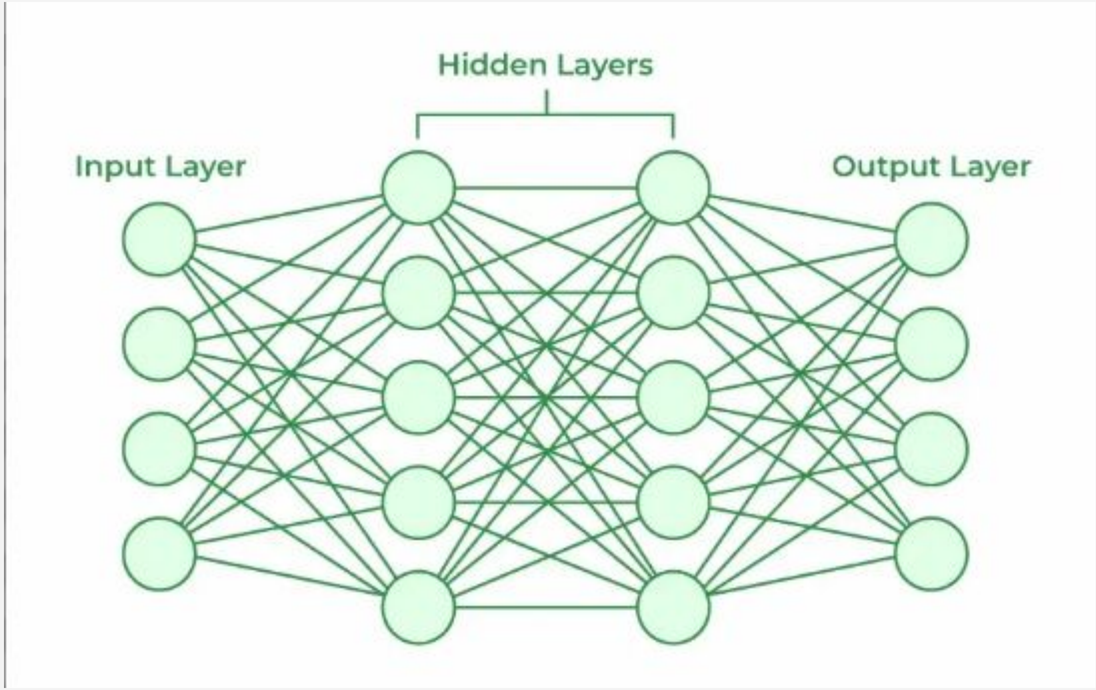
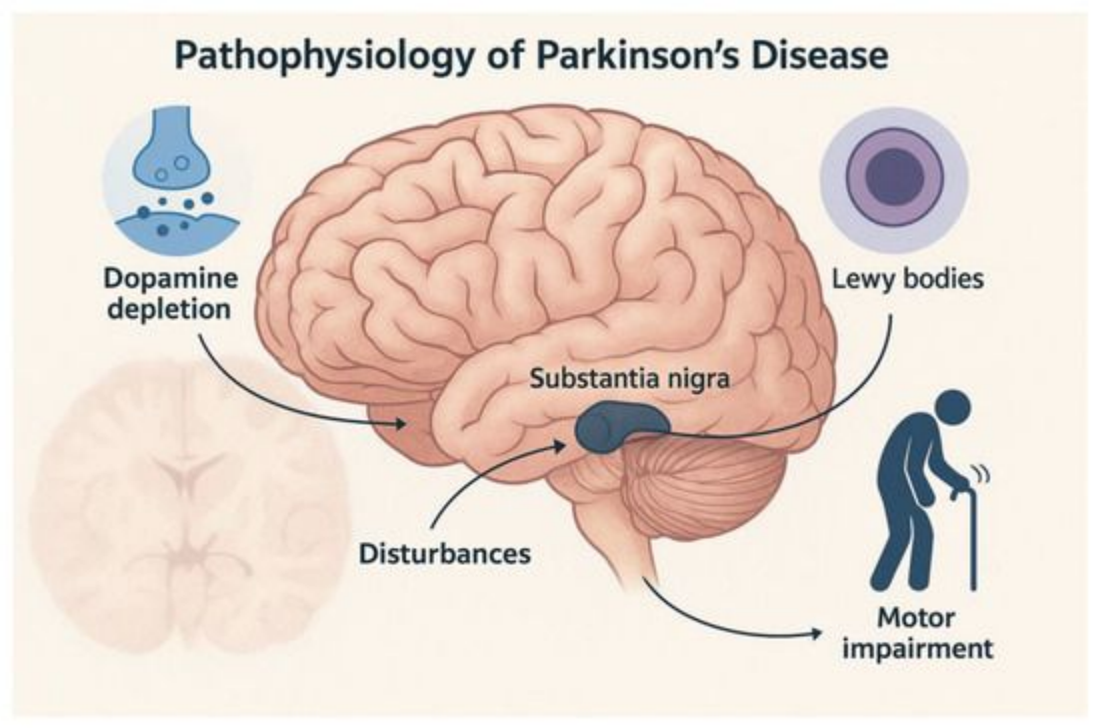
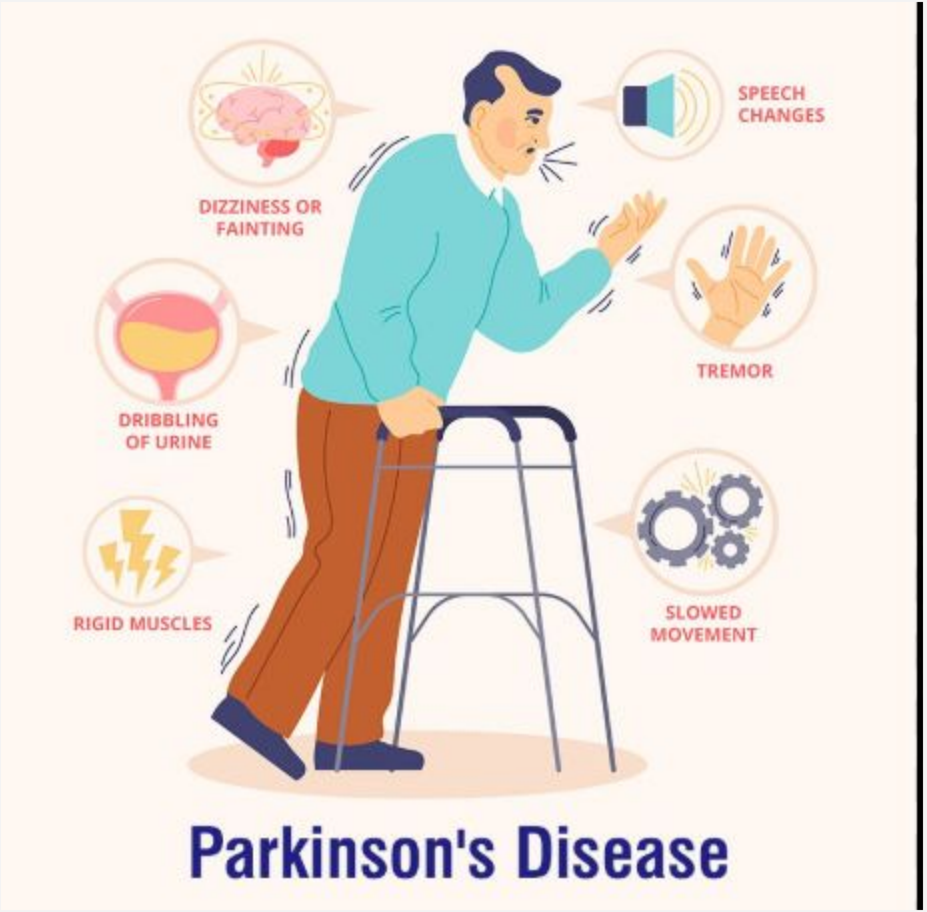
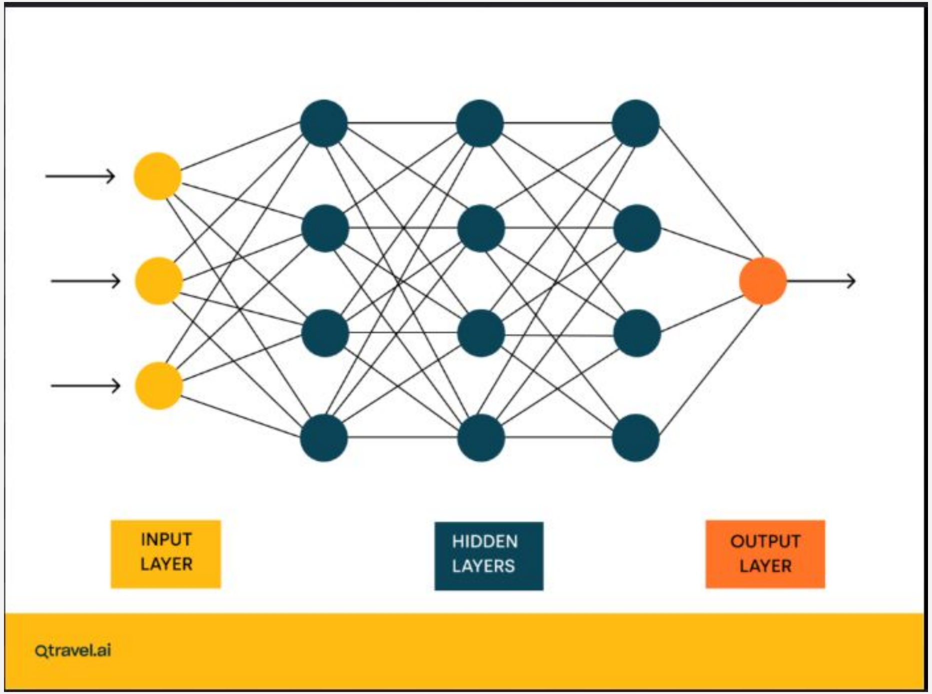
Background



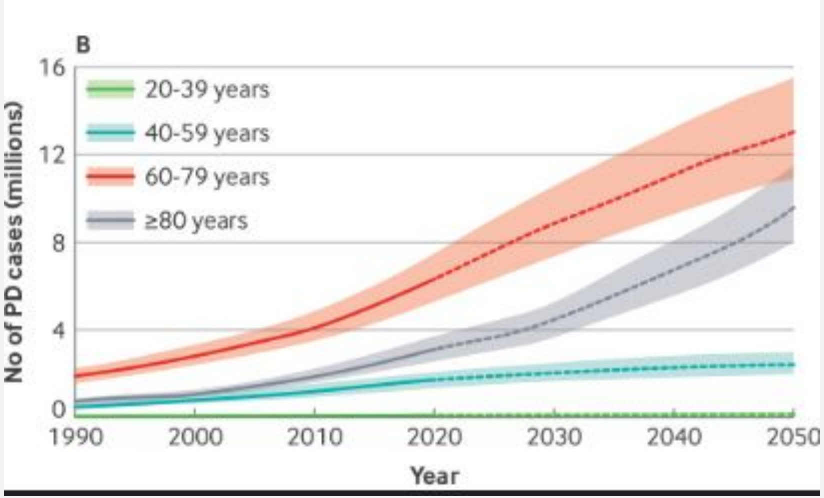

Pathophysiology of Parkinson's Disease
Dopamine depletion
Lewy bodies
Substantia nigra
Disturbances
Motor impairment

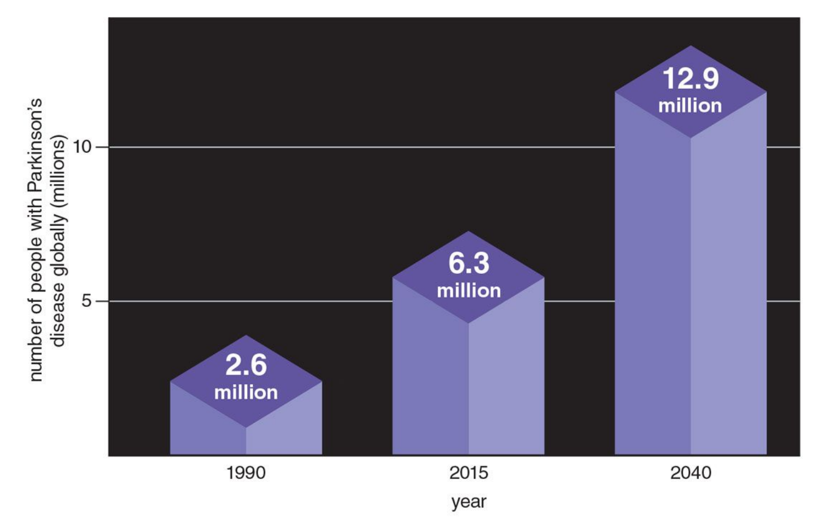

DIZZINESS OR FAINTING
SPEECH CHANGES
DRIBBLING OF URINE
TREMOR
RIGID MUSCLES
SLOWED MOVEMENT
Parkinson's Disease

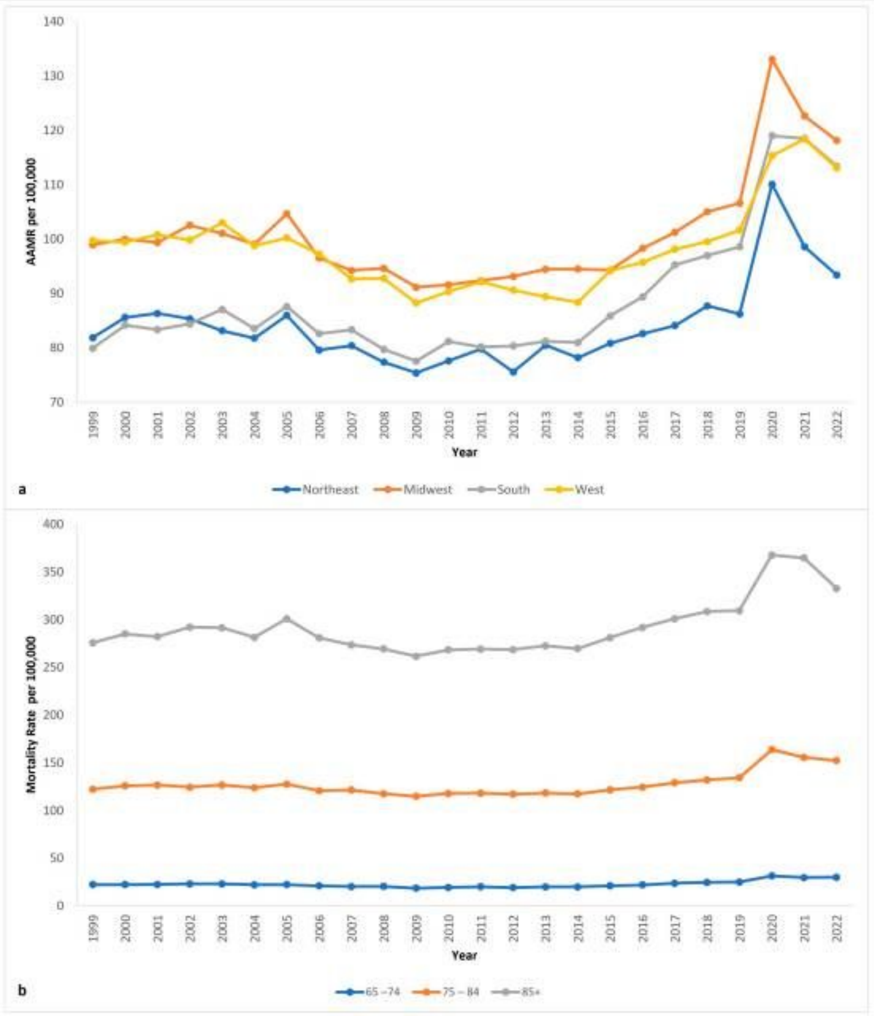
How might Artificial Intelligence be used to augment regenerative treatments to cure Parkinson's?



Parkinson's is the fastest growing neurological disease





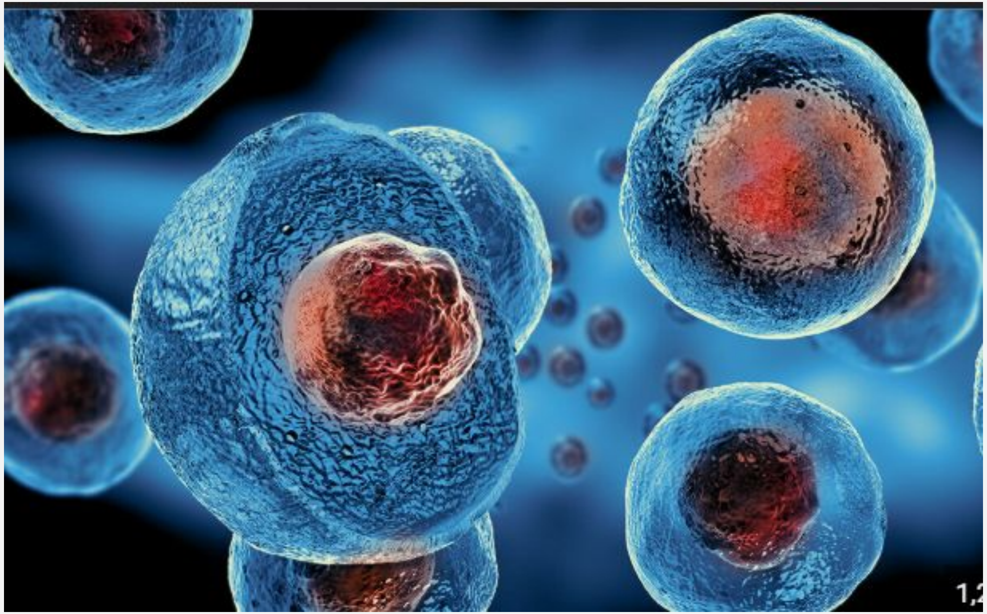


AAMR per 100,000
Year
a
Northeast
Midwest
South
West
Mortality Rate per 100,000
Year
b
65 –74
75 – 84
85+

Possible Treatment

Stem Cell Therapy

The one with the most potential



Gene Therapy

Primary Research – Meta Analysis

Key Themes

- The ability of AI to predict the outcomes of gene therapy procedures based on training data
- The ability of AI to predict the outcomes of stem cell therapy development based on training data
- The environment of the AI's training and study and whether the model's accuracy could translate to a clinical setup
- The ethical and social implications associated with using AI to develop treatment for disease
- Other drawbacks regarding AI such as a lack of large, comprehensive datasets and the "black-box" problem

Primary Research

AI-Driven Quality Monitoring and Control in Stem Cell Cultures: A Comprehensive Review

Biotechnology Journal
Systems & Synthetic Biology · Nanobiotech · Medicine

REVIEW | Open Access

AI-Driven Quality Monitoring and Control in Stem Cell Cultures: A Comprehensive Review

Rohan Singh, Hamid Ebrahimi Orimi, Praveen Kumar Raju Pedabaliyarasimhuni, Corinne A. Hoesli, Moncef Chioua

First published: 10 August 2025 | https://doi.org/10.1002/biot.70100 | VIEW METRICS

Funding: This research was funded by Médicament Québec, project# RQM00158.

Primary Research

Listen

Perspective

Artificial intelligence powers regenerative medicine into predictive realm

Armin Garmany & Andre Terzic

Pages 611-616 | Received 25 Sep 2024, Accepted 29 Nov 2024, Published online: 11 Dec 2024

Cite this article https://doi.org/10.1080/17460751.2024.2437281 Check for updates

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Primary Research

nature biotechnology

Article | Published: 16 January 2023

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Nicolas Mathis, Ahmed Allam, Lucas Kissling, Kim Fabiano Marquart, Lukas Schmidheini, Cristina Solari, Zsolt Balázs, Michael Krauthammer & Gerald Schwank

Nature Biotechnology **41**, 1151–1159 (2023) | Cite this article

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Article | Published: 10 December 2024

Artificial intelligence-guided design of lipid nanoparticles for pulmonary gene therapy

Jacob Witten, Idris Raji, Rajith S. Manan, Emily Beyer, Sandra Bartlett, Yinghua Tang, Mehrnoosh Ebadi, Junying Lei, Dien Nguyen, Favour Oladimeji, Allen Yujie Jiang, Elise MacDonald, Yizong Hu, Haseeb Mughal, Ava Self, Evan Collins, Ziying Yan, John F. Engelhardt, Robert Langer & Daniel G. Anderson

Nature Biotechnology **43**, 1790–1799 (2025) | Cite this article

Key Takeaways

- AI is super accurate
- Datasets
- Potential for bias
- The Black-box problem
- Data Privacy
- No Human Trials

"The longer I had the disease and was moving around the planet, I realized people were just waiting for this ethereal cure to happen.... And I fairly quickly realized that waiting was too passive."

- Davis Phinney

Resources

Data

- Su, Dongning, et al. "Projections for prevalence of Parkinson’s disease and its driving factors in 195 countries and territories to 2050: modelling study of Global Burden of Disease Study 2021." bmj 388 (2025).
- Jeong, S.H., Hoon Kim, S., Lee, D.W. et al. Association between new-onset Parkinson’s disease and suicide risk in South Korea: a nationwide cohort study. BMC Psychiatry 22, 341 (2022). https://doi.org/10.1186/s12888-022-03990-4
- Weresh, Henry et al. “Trends and Disparities in Parkinson's Disease Mortality in the United States with Predictions Using Machine Learning.” NeuroSci vol. 6,1 6. 15 Jan. 2025, doi:10.3390/neurosci6010006

Statistics

https://www.parkinson.org/understanding-parkinsons/statistics#:~:text=💡%20Quick%20Summary,90%2C000

Lindvall, Olle. “Treatment of Parkinson’s Disease Using Cell Transplantation.” Philosophical Transactions: Biological Sciences, vol. 370, no. 1680, 2015, pp. 1–7. JSTOR, http://www.jstor.org/stable/24505278

https://www.americanscientist.org/article/the-rise-of-parkinsons-disease

Primary Research

- Garmany, Armin, and Andre Terzic. "Artificial Intelligence Powers Regenerative Medicine into Predictive Realm." Regenerative Medicine, vol. 19, no. 12, Dec. 2024, pp. 611–616. Taylor & Francis Online, https://doi.org/10.1080/17460751.2024.2437281.
- Mathis, Nicolas, et al. "Predicting Prime Editing Efficiency and Product Purity by Deep Learning." Nature Biotechnology, vol. 41, no. 8, Aug. 2023, pp. 1151–1159. Nature, https://doi.org/10.1038/s41587-022-01613-7.
- Singh, Rohan, et al. "AI-Driven Quality Monitoring and Control in Stem Cell Cultures: A Comprehensive Review." Biotechnology Journal, vol. 20, no. 8, Aug. 2025, e70100. Wiley Online Library, https://doi.org/10.1002/biot.70100.
- Witten, Jacob, et al. "Artificial Intelligence-Guided Design of Lipid Nanoparticles for Pulmonary Gene Therapy." Nature Biotechnology, vol. 43, 2025, pp. 1790–1799. Nature, https://doi.org/10.1038/s41587-024-02490-y.